

Beyond the Leaderboard: Counterfactual Diagnosis of End-to-End and VLA Driving Policies Under Domain Shift

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Abstract—End-to-end and vision-language-action (VLA) driving policies are compared by leaderboard rank, but a deployment decision needs to know how a policy will behave at a new site, and why.

Attempting to predict a policy’s performance at a new deployment site from an existing leaderboard therefore works poorly, because a rank reflects only the outcome, not the behaviour behind it. On six released policies, rank on nuScenes open-loop metrics or on NAVSIM’s leaderboard does not carry over to scenes with a pedestrian near the ego corridor at a new site (from -0.54 to $+0.49$).

We propose a counterfactual check-up: a few hundred real frames, each edited two ways (pedestrian removed; night-style re-rendering), every edit verified by an independent detector, and the change in the planned trajectory read as a diagnosis rather than a score. By distilling causal feature axes from these edits, five exams separate what a score merges: how far the policy plans to drive, whether seeing the pedestrian buys safety, whether that response scales with danger, whether the plan moves when nothing requires it, and how much an irrelevant lighting change moves it.

Across 246 near-pedestrian scenes and six policies, only 2.4% of responses are genuine avoidance, while a near-zero response reliably certifies that the pedestrian was ignored (93–99% of scenes). In a pre-registered test from left- to right-hand drive, orderings and verdicts transfer but point values do not, and the exam that best predicts the new side’s near-miss rate is how steadily a policy drives ($= -0.94$), not how strongly it reacts. Most verdicts settle within a few dozen frames, while hazard sensitivity needs hundreds. Code and edited frames will be released.

I. Introduction

A leaderboard rank does not tell you how a driving policy will behave in a new domain. End-to-end and vision-language-action (VLA) driving policies are compared by benchmark scores, and a deployment decision is, implicitly, a bet that the rank those scores produce will still hold once the policy is driving at the new site, not just on someone else’s roads. On scenes where a pedestrian is near the ego corridor, that bet fails for the six released policies we test: their rank on nuScenes’ open-loop metrics or on NAVSIM’s leaderboard score does not carry over to the deployment site ($= -0.54$ to $+0.49$; Section V). The reason is not noise but mechanism. A score is an outcome measure, and an outcome can be right for the wrong reason: a policy that never approaches a pedestrian because it plans short trajectories scores as well as one that sees the pedestrian and yields, and the two behave very differently the first

time the scene changes. Telling them apart means asking why the score came out as it did, usually from little data, because a closed-loop simulator for the target domain or a fleet test is exactly what the decision is meant to save.

This paper shows that a few hundred edited frames do tell you, and reports what transfers to a new domain and what does not. We propose a check-up: a short list of exams, each with a control and a reference range, whose results are read as a diagnosis rather than a ranking (Table I). Every exam is a counterfactual edit of real frames. We remove the pedestrian from the ego corridor, or re-render the frame with a night-style photometric perturbation, and read the change in the planned trajectory; an independent detector verifies every edit. The central exam is a hazard-sensitivity curve, measuring how much farther the whole planned trajectory stays from the pedestrian’s logged future because the policy can see the pedestrian, against a model-independent hazard level. A specificity term applies a minimal-change principle: when no reaction is needed, the plan should not move.

A deployment decision is, implicitly, a bet that a benchmark rank still holds at the new site. The bet fails for a reason that is mechanical rather than statistical: a score is an outcome measure, and an outcome can be right for the wrong reason. A policy that never approaches a pedestrian because it plans short trajectories scores as well as one that sees the pedestrian and yields, and the two behave very differently the first time the scene changes. Telling them apart means asking why the score came out as it did, usually from little data: a closed-loop simulator for the target domain, or a fleet test, is exactly what the decision is meant to save.

We frame this as a transfer problem from the benchmark to the deployment domain. Current practice closes the gap empirically: a candidate policy is exercised at scale in the target domain, adjusted and post-trained on the resulting feedback, and exercised again, the loop repeating until its behaviour is judged acceptable. Every turn of that loop consumes substantial testing resources and lengthens the development cycle.

A considerable body of cross-domain work has been proposed—domain adaptation, domain generalisation and the like—but it concentrates on identifying the problem and on solving it through training, and lacks any sys-

tematic assessment of how a policy behaves in the target domain. It therefore does not substantively address the selection problem that arises at deployment.

In Don't Blind Your VLA [?], a think-site representation is built to measure the degree of language-image segmentation; the transfer of that suite, however, applies only to vision-language multi-modal models and does not extend to the evaluation of decoder-only diffusion policies, and it is still built upon an already-chosen model, studying how that model's capability may be improved rather than answering systematically how a model should be selected.

In the RepE paper [1] the authors give a method for locating and repairing the gap between domains, but it is aimed at exposing a large language model's response to semantic stimuli, and is likewise hard to extend to the evaluation of physical-AI policies that read images.

Further work on measuring transfer in large language models, such as function vectors [?], shows that transfer evaluation is already established there; these methods, however, remain difficult to apply directly to the cross-domain evaluation of physical-AI policies. A systematic method for the cross-domain evaluation of physical-AI policies is still missing.

The consequence of this gap in the physical-AI literature is that the decision is left to the leaderboards. In autonomous driving a growing number of leaderboards and open test sets have appeared [2]–[4], meant to narrow the gap between the conditions a model is trained on and those it is deployed in, and with them a variety of ratings and rankings. nuScenes' leaderboard and NAVSIM's leaderboard, for instance, give researchers a means of assessing how a policy performs under different conditions. The scores these leaderboards report, however, are for the most part conventional traffic metrics. With the recent proliferation of large-model-based methods, reading the final outcome alone no longer reflects a policy's capability in full [5]–[7]. !!!!!!!!!!! ADD MORE PAPER TO REFER

The consequence is that a policy ranked first on a leaderboard often falls sharply once the domain changes, or the ranking ceases to be informative at all [8], [9], so that it cannot serve as a sound basis for the selection decision that takes a policy from simulation into a real deployment domain.

To address this issue, we propose a check-up: checkup

We construct pixel-level paired data and reduce each pair to two axes of causal attribution: the faithfulness displacement F , the plan change caused by the pedestrian itself, and the interference displacement I , the change the same policy produces under an edit that carries no driving-relevant information (3). Reading a policy on these axes yields a prognosis for its behaviour in the target domain: diagnosed on one driving side and tested on the other, the exam that transfers predicts the new side's near-miss rate at $\rho = -0.94$, whereas nuScenes' open-loop metrics and NAVSIM's leaderboard score reach only -0.54 – $+0.49$

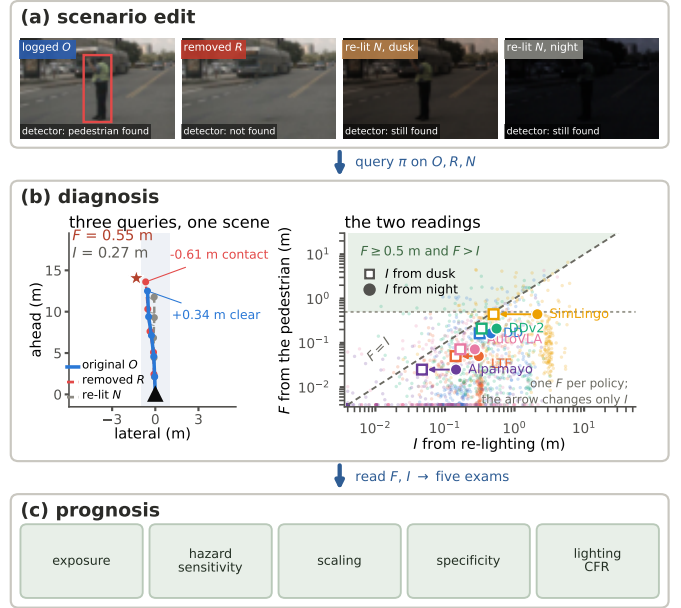


Fig. 1. check-up frame. The check-up proceeds in three stages: constructing pixel-level paired data, distilling the causal axes from them, and issuing a prognosis for the policy's behaviour in the target domain.

against the same target; what carries over is the ordering and the verdict. Fitting how each estimate's spread shrinks with sample size further shows the lighting verdict settled within 14 frames and exposure within 51, so a deployment decision can be priced in tens of target-domain frames rather than in a closed-loop simulator or a fleet test. The framework therefore offers a more dependable basis for selecting a policy for a given deployment.

The contribution of this paper is a small-sample check-up for end-to-end and VLA driving policies, built on verified, pixel-level counterfactual edits of real frames. It comprises

- 1) the construction of pixel-level paired data: built a small-sample check-up for driving policies on verified, pixel-level counterfactual edits of real frames rather than the promptable text prompts language-model interpretability usually relies on — to our knowledge among the first uses of concept-vector-style methods on a multi-modal, largely non-promptable input. Its central result is methodological: raw response magnitude is a poor safety signal, and decomposing it into magnitude, jitter and separation turns a two-sided quantity into a one-sided certificate. We use it to diagnose six policies, with an explanation from inside the networks.
- 2) diagnose with the distillation of the causal axes: we treat the raw response to removing a pedestrian as a poor safety signal, because a plan that moves is as likely to move towards the pedestrian as away from it. Decomposing the response into magnitude,

which is represented as the faithfulness direction $\hat{v}_F$; and the policy’s own jitter under an irrelevant edit, which is the interference direction $\hat{v}_I$. And the separation it buys turns a two-sided quantity into a one-sided certificate: a near-zero response shows the pedestrian did not enter the decision; a large one shows nothing. A diagnosis with an explanation from inside the three BEV planners of why the pedestrian risk goes unmanaged: the pedestrian leaves a trace in the features but is not decoded, the planning grid cannot locate it, and the plan rarely reaches it.

- 3) the prognosis: An account of what a small check-up can and cannot establish in a new domain: a sample-size law for every exam, and a pre-registered transfer test from left- to right-hand drive. Orderings and verdicts transfer; point values do not. The exam that predicts behaviour on the new side is how steadily a policy drives, not how strongly it reacts, and it does so better than that side’s own leaderboard score.

II. Related Work

a) Driving policies and their benchmarks: We examine bird’s-eye-view (BEV) planners of the TransFuser lineage [10] (LTF, DiffusionDrive [11], DiffusionDriveV2 [12]) and vision–language–action models (SimLingo [13], AutoVLA [14], Alpamayo-1.5 [15], [16]). NAVSIM [4], [17] scores their plans by multiplying hard gates with a weighted average of soft terms. Open-loop scores are known to reward proxies [5]. We show that the six-policy spread is carried by a single gate.

b) Behavioural testing: Checklists of targeted behavioural tests [18] and corruption benchmarks [8], [9], [19] probe models beyond aggregate accuracy. A metric drop, however, does not say whether a policy became more cautious or more reckless. Our exams read signed plan changes against a counterfactual and state how many frames each needs.

c) Causal confusion: Imitation learners can respond to correlates of the expert’s action rather than to its causes [6], [7], [20], [21]. Our exams apply this idea to trained policies: a cause (a pedestrian) and a nuisance (lighting) are intervened on directly.

d) Counterfactual editing and internal representations: Counterfactual explanations edit the scene [22], [23]; we verify every edit with a detector. Steering directions [1] can show that a network encodes what it does not act on [24]; we use them to explain a diagnosis, not as the diagnosis.

III. The Check-Up

A. Units, interventions and plans

Write a policy as a map π from a sensory-and-state input to a planned trajectory $X = (X_t)_{t \in [0, 2.5]}$, $X_t \in \mathbb{R}^2$ in the ego frame, sampled at 0.1 s. A unit is a pair $u = (f, v)$: a logged frame f and an input ego speed v . We take v from the log for every frame and, wherever

TABLE I

The check-up. Each exam, the question it asks, the counterfactual that answers it, the paired plans or axes of Section III it is built on, and the normal range of a driver that yields.

Exam	Question	Measurement	Built on	Normal
Exposure	How far ahead the plan commits	planned / logged distance, 2.5 s	X^O, H	≈ 1
Hazard sensitivity HS	Does it react to danger	safety gained by seeing the pedestrian, when needed	X^O, X^R	rises to 1 with hazard
Scaling	Does caution scale with danger	rank corr. of HS with hazard	HS vs a_{req}	> 0
Specificity SP	Does it over-react	plan change when no reaction is needed	F	≈ 1
Lighting CFR, align	Does it react to noise	perturbation vs. removal, same frame	F/I	CFR $\gg 1$, align ≈ 0

the pedestrian’s logged future is available, additionally $v \in \{4, 8\}$ m/s, which places the pedestrian on the planned path in scenes where the logged ego was stationary. The same grid is used on both corpora so that all six policies are read on one unit pool; the readouts are insensitive to it (widening the grid to $\{2, 4, 6, 8\}$ leaves the specificity and scaling orderings identical, $\rho = 1.00$).

Three interventions act on the input, leaving the policy untouched:

$$\mathcal{E}_O = \text{id}, \quad \mathcal{E}_R : \text{delete the pedestrian}, \quad \mathcal{E}_N : \text{re-light}, \quad (1)$$

and yield the paired plans $X^O = \pi(\mathcal{E}_O u)$, $X^R = \pi(\mathcal{E}_R u)$, $X^N = \pi(\mathcal{E}_N u)$. All quantities below are functionals of this triple and of the pedestrian’s logged future F_t . The horizon 2.5 s is the one all six policies share.

The displacement between two plans is

$$\bar{D}(X, Y) = \frac{1}{|\mathcal{T}|} \sum_{t \in \mathcal{T}} \|X_t - Y_t\|, \quad \mathcal{T} = \{0, 0.1, \dots, 2.5\}, \quad (2)$$

and the two quantities the rest of the paper is written in are

$$F = \bar{D}(X^O, X^R), \quad I = \bar{D}(X^O, X^N). \quad (3)$$

We call F the faithfulness displacement—the plan change attributable to the pedestrian itself—and I the interference displacement—the change the same policy produces under an edit that carries no driving-relevant information, and therefore its own jitter floor. A policy that uses the pedestrian and ignores the lighting has $F \gg I$; the reverse ordering means the plan is driven by something the road does not care about. Section V reads the same two contrasts in representation space, as unit directions $\hat{v}_F$ and $\hat{v}_I$.

a) Realising $\mathcal{E}_R$: The target is localised by projecting its annotated 3D box into the camera rather than by

taking the dataset’s 2D boxes, whose association to the tracked object is unreliable for the short-range, partially occluded pedestrians of these scenes. It is segmented with SAM [25] (box and torso prompts, clipped to the box, dilated by 9 px) and inpainted with LaMa [26]; for the LiDAR policy the points inside the box are deleted. Vision–language–action (VLA) policies read several cameras and only the front one is edited, so we retain only frames in which the target appears in no other view.

b) Realising $\mathcal{E}_N$: Each image $\mathbf{I} \in [0, 1]^{H \times W \times 3}$ the policy consumes is mapped by a closed-form photometric transform

$$\mathcal{E}_N(\mathbf{I}) = \text{clip}[g(\text{sat}_s \mathbf{I})^\gamma \odot \boldsymbol{\tau} + \sigma \boldsymbol{\epsilon}], \quad \boldsymbol{\epsilon} \sim \mathcal{N}(0, 1), \quad (4)$$

with sat_s the saturation scaling, $\boldsymbol{\tau}$ a per-channel tint and the noise seed fixed per frame. This is a photometric night-style perturbation (gamma, gain, saturation, noise), not a captured night scene; we refer to it as the night-style perturbation. Two levels are used: the milder dusk $(\gamma, g, s, \sigma) = (1.45, 0.70, 0.80, 2/255)$ with a warm tint, chosen to keep the pedestrian visible, and night $(2.2, 0.45, 0.45, 7/255)$ with a blue tint (Section V).

c) Validity of the interventions: An independent detector (Faster R-CNN [27], COCO, score ≥ 0.5) is run on the 248 front-only frames. It recovers the target in 100% of the original images, in 8.5% under $\mathcal{E}_R$ and in 89.9% under $\mathcal{E}_N$ (IoU ≥ 0.5): $\mathcal{E}_R$ suppresses the pedestrian for an observer outside the policy, while $\mathcal{E}_N$ leaves it visible and therefore changes the lighting rather than the scene. Two controls bound the residual artefact: pasting inpainting patches on empty road puts it at 5–17% of the removal effect at pedestrian scale, and removing off-road pedestrians, which should not matter causally, gives a same-type null.

B. Readouts

With $c_t(X) = \|X_t - F_t\| - r_e - r_p$ the clearance net of the ego half-width $r_e = 1.0$ m and the pedestrian radius $r_p = 0.4$ m, a plan is summarised by four functionals,

$$S(X) = \text{mean}_t \min(c_t, \bar{c}), \quad A(X) = \mathbb{1}[\min_t c_t > 0], \quad (5)$$

$$C(X) = \text{clip}(\min_t c_t / d_0, 0, 1), \quad T(X) = \frac{1}{2.5} \inf\{t: c_t < d_0\}, \quad (6)$$

with $T = 1$ if the set is empty. The constants are physical, not tuned: r_e is the half-width of the nuScenes ego box (4.08×1.85 m), r_p the disc a standing adult occupies, $d_0 = 1$ m the outer edge of personal space [28], and $\bar{c} = 10$ m caps the contribution of scenes in which the pedestrian is far away. Equations (5)–(6) instantiate the standard surrogate-safety measures—contact, minimum clearance and an encroachment-time variant [29], [30]—read as distance to a person over the whole trajectory, as is conventional for navigation near people [31]. Speed is deliberately not a readout: the same deceleration is safe or unsafe depending on where the person is.

C. Exams

Let $L(X) = \sum_t \|X_{t+\delta} - X_t\|$ denote arc length and H the logged human trajectory.

a) Exposure: $\text{Exp} = \text{med } L(X^O)/L(H)$ over frames with $L(H) \geq 0.5$ m.

b) Hazard sensitivity: A unit needs a reaction when the pedestrian-blind plan enters personal space, $C(X^R) < 1$. On those units, writing $\Delta(\cdot) = (\cdot)(X^O) - (\cdot)(X^R)$,

$$\text{HS} = \frac{1}{4} [\Delta A + \Delta C + \Delta T + \tanh(\Delta S/d_0)] \in [-1, 1]. \quad (7)$$

Each term lies in $[-1, 1]$ by construction and $\tanh$ squashes the single unbounded one at the d_0 scale. The composite is ours; its ingredients are not. Its four terms are the surrogate-safety measures of (5)–(6), and averaging normalised sub-scores into one index is how the benchmark we compare against forms its extended predictive driver model score (EPDMS) [4], keeping the two commensurable. What the check-up adds is that every sub-score is a paired difference across $\mathcal{E}_R$ —the ablation form used to attribute behaviour to an input in interpretability work [1], [32]. HS is read against the model-independent hazard

$$a_{\text{req}} = \frac{v^2}{2 \max(d - 2, 0.5)}, \quad \text{TTC}_0 = \frac{d - 2}{v}, \quad (8)$$

the deceleration required to stop short of a pedestrian at distance d and the corresponding time to collision (TTC) [29], [30]. Units with $v < 1$ m/s are excluded: a stationary ego can move neither toward nor away from anyone.

c) Scaling and specificity: $\text{Sc} = \rho(\text{HS}, a_{\text{req}})$ over needed units, with ρ Spearman’s rank correlation. On units needing no reaction, the minimal-change principle gives

$$\text{SP} = \exp(-\bar{D}(X^O, X^R)/0.5 \text{ m}) \in (0, 1], \quad (9)$$

so $\text{SP} \rightarrow 1$ iff an irrelevant edit leaves the plan where it was.

d) Lighting: The causal faithfulness ratio weighs the two displacements of (3) against each other,

$$\text{CFR} = \frac{\mathbb{E} \bar{D}(X^O, X^R)}{\mathbb{E} \bar{D}(X^O, X^N)} = \frac{\mathbb{E} F}{\mathbb{E} I}, \quad (10)$$

$$\text{align} = \text{med } \cos(X^N - X^O, X^R - X^O). \quad (11)$$

A policy for which lighting is irrelevant has $\text{CFR} \gg 1$ and $\text{align} \approx 0$; positive alignment means the re-lighting pushes the plan the way deleting the pedestrian does, and align is the cosine of the two representation-space directions of Section V.

D. Policies and corpora

We examine LTF and DiffusionDrive (front camera and ego status), DiffusionDriveV2 (camera and LiDAR), SimLingo (InternVL2-1B), AutoVLA (Qwen2.5-VL-3B; three cameras $\times$ four frames) and Alpamayo-1.5 (four

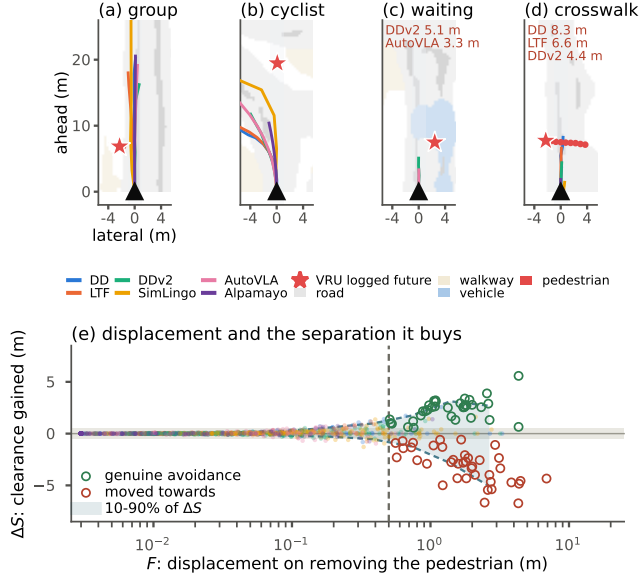


Fig. 2. Four right-hand-drive cases, and displacement against the separation it buys. (a–d) Ego at the origin, the vulnerable road user’s (VRU) logged future dashed from the star, and DiffusionDriveV2’s own BEV semantic head as the background; every policy receives the frame’s logged navigation, so the turns are commanded. (a) Clearances 0.3–0.9 m. (b) SimLingo passes at 3.2 m, the other five at 6.7–10 m. (c, d) The departing policies and their planned distances are named in the panel; Boston departure rates are DiffusionDriveV2 81%, AutoVLA 9%, DiffusionDrive 2.8%. (e) Every policy–scene cell of the 246 Singapore scenes: F against ΔS ; band: the 10–90% spread of ΔS per F bin; dashed line: $F = 0.5$ m.

cameras; a diffusion head over the vision–language model (VLM) cache), all with released weights. DiffusionDrive’s diffusion seed is fixed so that paired queries share their noise.

The primary corpus is nuScenes [2] keyframes: corridor pedestrians (lateral clearance to the ego corridor ≤ 1 m) at 5–12 m, their off-road counterparts, and 30 approach sequences at 5.8–14.6 m with logged pedestrian futures. The near-pedestrian units come from 236 front-only frames, 88 in Boston (left-hand drive) and 148 in Singapore (right-hand drive); Alpamayo-1.5 covers 170 of the 236, needing 1.6 s of history and 6.4 s of future within the scene. Section IV repeats $\mathcal{E}_R$ and $\mathcal{E}_N$ on a second, right-hand-drive corpus—the NAVSIM test split [4], 254 Singapore scenes with a pedestrian near the ego corridor, of which 246 carry a complete logged pedestrian future—used only for the transfer and decomposition results; Table III is measured on the nuScenes frames alone.

IV. Diagnosis: Reading the Two Axes

The diagnosis reads a policy on the two axes of (3). We apply both interventions to the 246 NAVSIM Singapore near-pedestrian scenes and read, per scene, F , the plan displacement caused by removing the target pedestrian, and I , that caused by the night-style perturbation.

TABLE II

What the night-style perturbation does to the plan. CFR: plan change from removing the pedestrian over that from the night-style perturbation, same frames (95% CI; ideal $\gg 1$). Speed: change of planned mean speed under the perturbation. ΔS : change of whole-plan separation from the pedestrian (negative = closer). Align: direction of the perturbation-induced change relative to the removal-induced one. Bold: $p < 0.01$.

Policy	CFR [95% CI]	speed	ΔS (m)	align
DD	0.30 [0.20,0.43]	+37%	−0.37	+0.55 → +0.09
LTF	0.36 [0.24,0.49]	+12%	−0.24	−0.09 → −0.42
DDv2	0.49 [0.33,0.70]	−8%	+0.15	+0.63 → +0.16
SimLingo	0.31 [0.21,0.44]	+16%	−0.47	+0.30 → +0.14
AutoVLA	0.38 [0.21,0.57]	−0.3%	+0.005	+0.65 → +0.62
Alpamayo 1.5	0.18 [0.10,0.30]	−2%	+0.03	+0.81 → −0.30

a) When the plan does move, which way does it move?: F measures how far a plan moves, not where it moves to. We therefore decompose it (Fig. 2e), reading F against I of (3)—the same policy’s jitter floor under an unrelated edit—and against ΔS , the change in clearance to the pedestrian’s logged future. A response counts as avoidance only if all three hold: the plan moved ($F \geq 0.5$ m), it moved more than the policy’s own jitter ($F > I$), and the movement bought distance ($\Delta S \geq 0.5$ m).

Three readings follow. First, the pedestrian outweighs the irrelevant edit in only 4–23% of scenes, so most of what F registers sits inside the policy’s own variability—the pinched band of Fig. 2e. Second, avoidance is rare where it is called for. The logged ego is stationary through most of this corpus, so a plan that does not move is usually the correct one; we therefore replay every scene at input speeds of 4 and 8 m/s, as in Section III, and read only the cells in which the pedestrian-blind plan does come within 1 m of the pedestrian’s logged future. Across 5 policies that is 569 of 3690 cells (11–24% per policy), and 15 of them—2.6%—are genuine avoidance. On those cells the median clearance changes by +0.00–+0.03 m and the median planned distance by −0.02–+0.08 m: facing a pedestrian it would otherwise reach, no policy diverts and none slows. Third, the displacement that does occur is not aimed: of the 121 cells in which the plan moved, 43 increased the clearance and 42 decreased it. Per policy the split differs sharply: DiffusionDrive reacts in few scenes but in the right direction (median $\Delta S = +2.16$ m), while SimLingo moves in more scenes than the other five combined and moves towards the pedestrian more often than away (32 against 13, median −0.59 m); it also has the highest rate of sub-1.5 s time-to-proximity (Table V). Finally, what a policy yields does not grow with the hazard: the rank correlation between ΔS and the deceleration the scene demands is −0.37–+0.07.

An exam that rewarded responsiveness alone would therefore rank the least safe policy first.

TABLE III

The check-up report. Normal range in brackets (Table I); ↓/↑ = outside it, bold = best in column. Collision: contact with the pedestrian’s logged future at 8 m/s input speed, visible / removed.

Policy	exposure [0.8, 1.2]	HS [≥0.15]	scaling [>0]	SP [≥0.8]	CFR [>1]	coll. 8 m/s vis. / rm.
DiffusionDrive	0.50↓	0.06↓	-0.28↓	0.76↓	0.23↓	7.6 / 9.2
LTF	0.79↓	0.03↓	-0.13↓	0.78↓	0.23↓	9.9 / 10.7
DiffusionDriveV2	0.88	0.10↓	+0.23	0.65↓	0.47↓	3.1 / 6.1
SimLingo	2.19↑	0.00↓	-0.22↓	0.44↓	0.23↓	40.5 / 36.6
AutoVLA	1.01	0.02↓	-0.18↓	0.76↓	0.44↓	32.1 / 29.8
Alpamayo 1.5	1.13	0.01↓	-0.23↓	0.84	0.25↓	11.2 / 11.2

V. Lighting: Magnitude, Direction, and the Pedestrian

Lighting should not change what a driver does, so this exam asks how far a policy is moved by an irrelevant change. It is the most robust result in the check-up and the one whose direction is least stable.

a) Magnitude: Removing a pedestrian from the corridor moves every policy’s plan only 0.18–0.49 times as far as the night-style perturbation does, with every 95% interval below one (Table II); on detector-verified edits CFR = 0.30–0.64, and only Alpamayo-1.5’s interval (45 frames) reaches above one.

b) Two lighting levels: The perturbation is applied at two strengths, the milder dusk level chosen to keep the pedestrian visible. Of the 45 targets the detector finds in the original, it still finds 89% at dusk and 67% at the night level (removal: 2%). On the same 246 scenes the ratio is 0.34–0.87 at dusk and 0.15–0.38 at the night level: below one for every policy at both levels, with the pedestrian in plain view.

c) Direction: faster, and therefore closer: A policy that slows under the perturbation is doing something defensible; one that speeds up is not. Three policies plan to go faster under the night-style perturbation: DiffusionDrive by 37%, SimLingo by 16% and LTF by 12%, and their whole plans pass 0.24–0.47 m closer to the pedestrian. The approach is longitudinal: the further the plan’s endpoint moves forward, the closer it gets (rank correlation -0.73 to -0.96), while the endpoint shifts toward the pedestrian’s side no more often than chance. DiffusionDriveV2, the only policy that also reads LiDAR, slows (-8%) and keeps more distance; AutoVLA and Alpamayo-1.5 are unaffected.

d) Direction is not a stable trait: Read in representation space as unit directions $\hat{v}_F$ and $\hat{v}_I$ (the activation-side counterparts of (3)), the two contrasts oppose one another in aggregate (mean cosine -0.54 over the four policies with accessible activations), yet their alignment changes with the driving side and its ordering is unstable even at 130 frames (Section VII). The magnitude of the lighting effect is a trait a small check-up can measure; its direction is not.

VI. The Report: Five Exams on Six Policies

A. The report

Table III and Fig. 3b give the six profiles.

a) Exposure: The six policies do not even agree on how far to plan, and this alone will turn out to explain more of the collision picture below than perception does: exposure spans a factor of four, from DiffusionDrive at half the human’s distance to SimLingo at more than twice it, with AutoVLA and Alpamayo-1.5 near the human’s own.

b) Hazard sensitivity: No policy manages the pedestrian risk. On units that need a reaction, seeing the pedestrian buys 0.00–0.10 of safety on a $[-1, 1]$ scale. The benefit does not grow with the hazard: its rank correlation with a_{req} is negative for 5 policies, down to -0.28 for DiffusionDrive; DiffusionDriveV2 is the only exception ($+0.23$), and its benefit stays below 0.15. Replacing the hazard axis by the time-based TTC_0 , whose sign is reversed by construction, leaves the picture unchanged (-0.07 to $+0.32$, against -0.28 to $+0.23$). The benefit is also unreliable in direction: counting only cases where seeing the pedestrian changes the outcome, it turns a collision into a clean pass about as often as it turns a pass into a collision, for every policy (DiffusionDrive 9 vs. 5, SimLingo 18 vs. 30, the rest at most 5 either way; all McNemar $p \geq 0.11$) — what reads as a safety margin in Table III is, case by case, close to a coin flip.

c) Specificity: A policy that moves when nothing requires it is not merely cautious, it is unpredictable, and unpredictability is what turns out to travel across a domain shift (Section VII). SimLingo’s plan moves most when nothing requires it ($\text{SP} = 0.44$); the other five stay steadier, at 0.65–0.84.

d) Collisions: Collisions with the pedestrian’s logged future follow how far a policy plans to drive, not what it perceives. At the logged speed, the visible and removed rates differ by at most 0.4 points for every policy, and at 8 m/s by at most 3.8. SimLingo, which plans the longest trajectories, collides most. The low-exposure policies almost never reach the pedestrian. DiffusionDrive is safe because it stays short, AutoVLA and Alpamayo-1.5 cover a human-like distance without yielding, and SimLingo covers the most ground with the least stable plan.

B. Why the risk goes unmanaged

The check-up only diagnoses; two mechanistic observations on the three BEV planners explain the diagnosis without altering it. First, the planners’ own BEV semantic heads do not decode the pedestrian: on 520 NAVSIM near-pedestrian scenes the pedestrian channel is no brighter at the pedestrian than at a mirrored empty location, and never wins the argmax inside a pedestrian’s box, although the pedestrian does leave a steerable trace in the features. Second, the plan does not reach the pedestrian: plans end 5–9 m ahead while the median corridor pedestrian stands

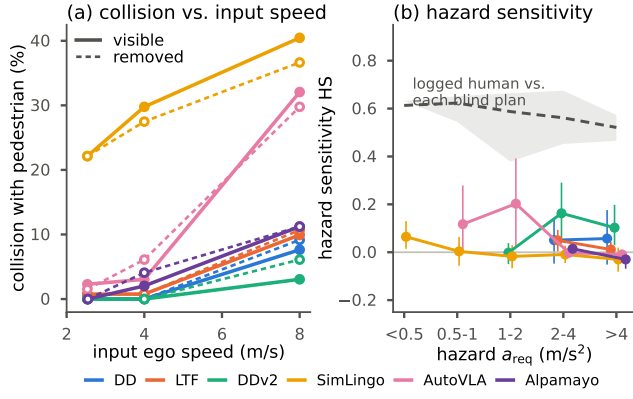


Fig. 3. Collision and hazard sensitivity. (a) Contact between the plan and the pedestrian’s logged future, at the logged speed and at counterfactual input speeds; solid: pedestrian visible, dashed: removed. (b) HS against the hazard level a_{req} (mean, 95% CI, bins with ≥ 8 units; points dodged within a bin, and a policy appears only where it has enough units). Dashed: the logged human trajectory scored against each policy’s own blind plan, averaged over policies; band: the range across the six.

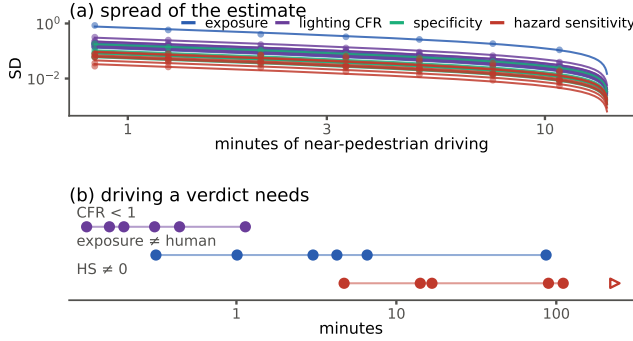


Fig. 4. How much driving a verdict costs. One frame is 5s of near-pedestrian driving. (a) Spread of each exam’s estimate against sample size, with its fitted law; one line per policy. (b) Driving needed to settle each verdict, one dot per policy; values beyond the axis are off scale.

20m away, and raising the input speed makes collisions rise identically with and without the pedestrian in view (Fig. 3a). The limit is geometric before it is perceptual.

VII. Prognosis: What Transfers, and at What Cost

A. Fitting

For every exam and policy we subsample n of the 170 frames that all six policies share, 400 times, and fit the spread of the estimate with $\sigma(n) = a\sqrt{1/n - 1/N}$ (Fig. 4).

a) Well-fitting exams: Exposure, the lighting ratio and specificity obey this law ($R^2 = 0.97-1.00$). The fitted a converts directly into the number of frames a verdict needs.

- Lighting outweighs the pedestrian (CFR < 1) is settled for every policy within 14 frames.

- Exposure differs from the human’s is settled within 51 frames for the four policies whose exposure is far from one. For AutoVLA and Alpaymayo-1.5 it would take hundreds to thousands of frames; this is itself a finding, since they plan a human-like distance.

- Orderings of the six policies stabilise quickly: exposure with 10 frames (96% of subsamples reach $\rho \geq 0.6$ with the full ordering) and specificity with 40 (99%).

b) Poorly fitting exams: Hazard sensitivity and the direction of the lighting effect behave differently. Their fits are poor (R^2 as low as 0.26 and -0.09) wherever few units need a reaction. Deciding that $HS \neq 0$ needs 57–1,326 frames, except for SimLingo, whose estimate sits on the threshold, and the ordering reaches 84% stability only at 90 frames. The ordering by alignment is unstable even at 130 (72%).

In wall-clock terms, a 20s scene in this corpus yields about 4 usable near-pedestrian frames, so the lighting verdict costs roughly 3 scenes and the exposure verdict 13; deciding that hazard sensitivity differs from zero costs 331—about 110 minutes of logged driving that actually contains a pedestrian near the corridor.

The division is therefore clear. A few dozen frames diagnose planned distance, over-reaction and sensitivity to lighting. Whether a policy manages hazard, and in which direction lighting pushes it, needs hundreds of frames that contain a genuine need to react.

B. A pre-registered transfer test

Does a diagnosis made on one driving side describe the same policy on the other? Before computing any right-hand-drive statistic, we fixed seven predictions and their decision rules. They were derived from the profile of the 88 Boston frames, and we tested them on the near-pedestrian units of the 148 Singapore frames (Table IV). The registration covered the six policies public at that time. NVIDIA subsequently released Alpaymayo 1.5, which accepts navigation instructions, and we use it in place of Alpaymayo-R1 throughout this paper; because both driving sides of Alpaymayo 1.5 were measured together, its numbers do not satisfy the registration discipline, so the table is evaluated on the five policies that do. Retaining the replaced policy changes none of the seven verdicts.

a) What transferred: 4 of the seven predictions hold, and they are the orderings and verdicts. The night-style perturbation outweighs the pedestrian in every policy. Exposure keeps its order ($\rho = 0.90$), and so does over-reaction, with SimLingo the most unstable. SimLingo collides most, with the visible and removed rates within 3 points of each other.

b) What did not: Three predictions fail.

- Point values drift: only 60% of the model-exam cells stay inside their Boston interval.
- In Singapore, DiffusionDrive and DiffusionDriveV2 show a small but non-zero hazard sensitivity ($HS = 0.09$ and 0.08).

TABLE IV

Pre-registered transfer test. Predictions fixed on 88 Boston frames before any Singapore statistic was computed, then tested on 148 Singapore frames. Five policies; keeping the replaced sixth changes no verdict (Section VII).

Prediction	Holds	Evidence
P1 Lighting outweighs the pedestrian (CFR < 1) for every policy	yes	max upper CI 0.70
P3 Exposure ordering transfers ($\rho \geq 0.6$; SimLingo top, DD bottom)	yes	$\rho = 0.90$
P4 Over-reaction ordering transfers ($\rho \geq 0.6$; SimLingo lowest SP)	yes	$\rho = 0.60$
P7 SimLingo collides most; visible vs. removed within 3 pts	yes	gap 2.7 pts
P2 Hazard sensitivity stays below 0.15 (upper CI)	no	fails for DD, DDv2
P5 Point values stay within the left-hand CI ($\geq 70\%$ of 15 cells)	no	60%
P6 Purer policies (lower align) shift less	no	$\rho = -0.90$ (opposite)

- The “purity” hypothesis is reversed. We had predicted that policies whose lighting effect is more orthogonal to their pedestrian response would shift less between sides; the observed rank correlation is -0.90 . Alignment is itself unstable between the two sides (Section V).

c) Reading: A small check-up can therefore report an ordering and a verdict for use in a new domain, but not a calibrated value.

C. Transfer of the method

a) What a small sample in the new domain buys: We draw k scenes at random from the NAVSIM pool of Section IV, re-measure F and I on them and compare against the large-sample values, 200 draws per k . Ordering and level behave very differently. The ordering of how dangerously the policies behave—the share of scenes whose minimum time-to-proximity falls below 1.5 s—is recovered almost immediately: $\rho = +0.64$ at $k = 10$ and $\rho = +0.65$ at $k = 20$ ($\rho \geq 0.6$ in 70% of draws), rising to $\rho = +0.76$ at $k = 40$. The level is not: the same $k = 20$ sample misstates the violation rate by 13 points and CFR by 0.116; the rule hit-rates sit in between, at 3.6 points. Importing the Boston CFR values instead misstates the Singapore ones by 0.099, already worse than re-measuring $k = 40$ fresh scenes (0.078), while 80 scenes reach 0.054 and 160 0.027. Values measured elsewhere are worth about forty scenes measured here.

b) Which exam travels: Of the exams in the report, the one that predicts the new domain’s behaviour is not the one that measures how much a policy reacts, but the one that measures how steadily it drives. Correlating each Boston exam against the Singapore rate of sub-1.5 s time-to-proximity over the six policies, specificity gives $\rho = -0.94$ —the only value significant at the two-sided 5% level for $n=6$ —against $+0.77$ for exposure and $+0.37$ for CFR. Specificity asks how far a plan moves when

TABLE V

The same six policies under the standard scores. Left: nuScenes open-loop L2 on the 236 near-pedestrian frames, original / pedestrian removed. Middle: NAVSIM EPDMS on all 783 Singapore scenes and on the 260 with a pedestrian near the corridor. Right: share of moving-ego scenes with minimum pedestrian TTC < 1.5 s, by driving side, with the rank in parentheses. Bold: best in column.

Policy	L2 (m) orig. / rm.	EPDMS		TTC<1.5 s	
		all	near	LHD	RHD
DD	0.98 / 0.99	0.535 (4)	0.744 (2)	3.2 (1)	14.1 (1)
LTF	0.88 / 0.91	0.509 (5)	0.735 (3)	5.4 (2)	16.5 (2)
DDv2	1.35 / 1.31	0.652 (3)	0.367 (6)	18.3 (3)	22.4 (5)
SimLingo	3.28 / 3.35	0.359 (6)	0.583 (4)	61.3 (6)	56.5 (6)
AutoVLA	0.68 / 0.66	0.683 (1)	0.413 (5)	19.4 (4)	21.2 (4)
Alpamayo 1.5	0.77 / 0.72	0.663 (2)	0.760 (1)	25.4 (5)	20.0 (3)

an irrelevant part of the scene is edited; a policy that answers “not at all” drives steadily, and steadiness is what survives the change of side. (Exposure travels too: on the larger original-image set of Section VIII it reaches 0.94 against the same target. Specificity is the stronger predictor and the only one that clears significance here.) The decomposition above says why: displacement that is not aimed away from the pedestrian is not protection, and a policy producing a lot of it is dangerous on both sides.

VIII. Beyond the Leaderboard: What the Check-Up Predicts

These policies are usually compared by a leaderboard rank or an open-loop error. Running those scores and the check-up on the same six policies (Table V) lets us ask what each recovers that the other does not, and which of them predicts behaviour at a site it has not seen.

a) nuScenes open-loop metrics: On our 236 near-pedestrian frames, removing the pedestrian from the image changes none of the standard open-loop numbers: the L2 error by at most 0.07 m, the frontal collision rate by at most 1.7 points and the pedestrian collision rate by at most 3.0 (Table V shows the L2 pairs; the two collision rates behave the same way and are omitted). The metric cannot tell whether a policy saw the pedestrian. Its ordering follows how far a policy’s planned distance departs from the human’s (rank correlation $+0.66$ with $|\text{exposure} - 1|$): L2 rewards driving at the human’s speed. We exclude rear-end contacts from the collision count. The logged traffic does not react, so a policy that plans to drive slower than the human is scored as being hit from behind.

b) NAVSIM, sliced to near pedestrians: On all 783 Singapore scenes, AutoVLA, Alpamayo-1.5 and DiffusionDriveV2 score highest. On the 260 scenes with a pedestrian near the ego corridor, the ordering is unrelated to that one ($\rho = -0.09$). The two rank columns of Table V can be read side by side: the leaderboard’s first (AutoVLA) falls to fifth and its third (DiffusionDriveV2) to last, while its fourth and fifth (DiffusionDrive, LTF) rise to second and third. Whichever policy a leaderboard reader would

have picked, the near-pedestrian scenes would have picked another. In these scenes the no-collision gate carries 100% of the spread, while over all 783 scenes it carries 1% and the two map-compliance gates carry 95% (drivable area 76%, driving direction 19%): the leaderboard ranks these policies by map compliance, and none of the exams above is visible in it. Of our exams, specificity predicts the near-pedestrian ordering ($\rho = +0.66$), while hazard sensitivity does not (+0.14), because no policy has any.

c) Pedestrian TTC across driving sides: Following the plan against the pedestrian’s logged future, we count the moving-ego scenes in which the minimum time to collision falls below 1.5 s. Near-pedestrian scenes from the left-hand-drive cities (nuScenes Boston and NAVSIM Las Vegas, Boston, Pittsburgh) and from Singapore give orderings that agree at $\rho = +0.77$. Both orderings coincide with planned distance ($\rho = +1.00$ and $+0.77$ with exposure). They agree only partly with hazard sensitivity (+0.83, +0.49), which is near zero for every policy, and not with lighting invariance (+0.14, -0.43). NAVSIM’s own TTC term, which counts all objects over 4 s, agrees only weakly (0.31–0.54).

d) Which evaluation predicts behaviour in another domain?: This is the question a deployment decision has to answer. The target is the ordering of the six policies on the Singapore near-pedestrian scenes (original images), by TTC violations, clearance to the pedestrian, pedestrian collisions and EPDMS. Each predictor is an ordering obtained without that data. The standard evaluations do not predict it: nuScenes L2 and collision rates, and the NAVSIM leaderboard score, correlate with it at -0.54 to $+0.49$. The same pedestrian-centred readouts taken on Boston or on the US cities do predict it (0.83–1.00), and so does the check-up’s exposure exam (0.94). Near-pedestrian EPDMS is predicted best by near-pedestrian EPDMS elsewhere (0.89) and by specificity (0.77). Hazard sensitivity and lighting invariance measure something else, and they do not predict these orderings.

e) Reading: Where a standard metric measures what an exam measures, they agree (pedestrian collisions: $\rho = +0.94$; TTC and exposure). None shows whether the pedestrian changed the plan, and the full-set score hides the near-pedestrian scenes.

IX. Discussion

a) Using the report: Read one exam at a time, the report answers separate questions that a single score merges. Whether a policy is safe because it yields or because it stays short is answered by exposure together with hazard sensitivity: DiffusionDrive keeps its distance by planning short trajectories, which holds only as long as the trajectories stay short, whereas AutoVLA and Alpamayo-1.5 cover a human-like distance without the response to match. Whether a plan that moves is a plan that yields is answered by the decomposition of

Section IV: among the cells in which the pedestrian-blind plan does reach the pedestrian, only 2.6% meet all three conditions, and displacement alone would have ranked the least safe policy first. The exam that no current benchmark reports is scaling: whether the response grows when the hazard does. Five policies fail it, and the one that passes, DiffusionDriveV2, does so at a sensitivity below 0.15.

b) Limitations: The exams are open-loop. The approach sequences are replayed from logs and re-anchored to the logged pose. The night-style perturbation is photometric, not a captured night scene, and it also halves the number of other pedestrians the detector finds. The counterfactual pairs are constructed by editing, not captured, and so carry noise: 8.5% of removals leave a detectable residue, the inpainting artefact is 5–17% of the effect, and the VLA models can be edited only through their front camera. Alpamayo-1.5 covers 170 of the 236 frames. SimLingo plans 2.5 s; for NAVSIM’s 4 s we extrapolate at constant velocity (holding position instead gives EPDMS 0.47, still the lowest). Cross-policy correlations rest on six policies.

c) Future work: The check-up is designed as the first half of a benchmark-to-deployment pairing, and as one hazard type among several a deployment decision must cover: extending its exams to lead-vehicle braking and turning or intersection conflicts is the natural next scenario. Within the pedestrian scenario, the next step is to collect paired data rather than construct it: the same scene with and without a pedestrian, in daylight and at night. We then plan to validate the diagnoses against the same policies’ behaviour on a real vehicle. The sample-size laws above tell us how large that collection must be. A few dozen pairs suffice for planned distance, over-reaction and lighting; hundreds of pairs with a genuine need to react are needed for hazard sensitivity.

X. Conclusion

A benchmark score reports an outcome, not the behaviour behind it, so a rank cannot be trusted to hold at a new site. We proposed a check-up that reads that behaviour from a few hundred logged frames through verified, pixel-level counterfactual edits, reducing each scene to two axes of causal attribution—the faithfulness displacement F and the interference displacement I —on which five exams are built. Three results carry the paper. First, on the cells in which a pedestrian does lie on the planned path, only 2.6% of responses are genuine avoidance, and no policy either diverts or slows; a photometric night-style perturbation moves every plan further than the pedestrian does. Second, diagnosed on one driving side and tested on the other, the exam that transfers predicts the new side’s near-miss rate at $\rho = -0.94$, where the standard scores reach only -0.54 – $+0.49$; what carries over is the ordering and the verdict. Third, each verdict can be priced in advance—the lighting

verdict within 14 frames, exposure within 51—so that a deployment decision costs tens of target-domain frames rather than a fleet test.

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